Random Forest Model to Predict Net Promoter Score among Fictional Bank Complaints

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Table of contents

1 Abstract **1**

2 Introduction **2**

3 Methods **3**
3.1 Dataset . . . 3
3.2 Descriptive Statistics . . . 3
3.3 Exploratory Data Analysis . . . 5
3.4 Random Forest Model . . . 9
3.4.1 Random Forest with One-Hot Encoding . . . 9
3.5 Feature Importance . . . 10

4 Results **12**
4.1 Model Performance . . . 12
4.2 Feature Importance . . . 12

5 Discussion **12**

6 Appendix **13**

7 References **13**

1 Abstract

This study explores the predictors of customer satisfaction as measured by Net Promoter Scores (NPS) using a random forest regression approach applied to a dataset of fictional bank complaints. The analysis identified Days_To_Resolve as the most important predictor, highlighting the central role of timely resolution in shaping customer perceptions. Although the correlation between resolution time and NPS was modest, its prominence across models underscores its practical significance. Among categorical variables, Language emerged as the most influential, with French-language complaints associated with slightly lower NPS. These findings suggest that operational improvements that shorten complaint resolution times may meaningfully boost customer satisfaction and, in turn, drive business growth. Future work will incorporate sentiment analysis of complaint text to further enhance predictive accuracy.

```
import pandas as pd
import numpy as np
import seaborn as sns

from sklearn.preprocessing import StandardScaler, OneHotEncoder
```

```
from sklearn.ensemble import RandomForestRegressor
from sklearn.model_selection import train_test_split
from sklearn.metrics import mean_squared_error
from sklearn.pipeline import make_pipeline
from IPython.display import Markdown

sns.set_context('paper')
```

```
# RQ1: Can a random forest model predict the NPS response?
# What is the best RMSE a random forest model can produce?

# RQ2: Among product name, days to resolve, and origin which factors are
# the most important in predicting NPS Response?

# RQ3: Revisit RQ2 but with the addition of the sentiment score as a predictor.
```

```
df = pd.read_csv(
    (
        "https://raw.githubusercontent.com/adamrossnelson/"
        "confident/refs/heads/main/data/complaints-bank.csv"
    )
)[[
    "CustomerID", "Product", "Date", "Complaint", "Language",
    "Days_To_Resolve", "Origin", "NPS Response", "Name", "Address"
]]
```

2 Introduction

In an increasingly customer-driven marketplace, understanding and predicting consumer sentiment has become a cornerstone of effective marketing strategy. Among the many metrics used to gauge customer experience, the Net Promoter Score (NPS) has emerged as one of the most influential. Introduced by Fred Reichheld in the Harvard Business Review in 2003, NPS simplifies customer feedback into a single question, "How likely are you to recommend our company to a friend or colleague?" and transforms the answer into a powerful signal of loyalty, advocacy, and business potential (Reichheld, 2003). The score, calculated by subtracting the percentage of detractors (those giving a score of 0–6) from the percentage of promoters (scores of 9–10), provides a clear, interpretable indicator of customer satisfaction that has been widely adopted across industries.

What sets NPS apart from traditional customer satisfaction measures is its strong correlation with long-term financial performance. According to Reichheld and other researchers, companies with higher NPS tend to outperform their competitors in areas such as growth, retention, and profitability (Reichheld, 2003; Markey, Reichheld & Dullweber, 2020). This is because NPS captures not just satisfaction, but emotional commitment and future behavior, which are two factors critical to driving organic growth through word-of-mouth referrals. From a marketing perspective, this makes NPS both a diagnostic tool and a strategic compass. Businesses can segment their customer base, tailor messaging, refine service delivery, and even optimize product development by understanding which customers are promoters, passives, or detractors, and why. Furthermore, as consumers become more vocal and empowered through digital platforms, a company's ability to convert detractors into promoters is more important than ever for maintaining brand reputation and competitive advantage.

By leveraging data science techniques to predict NPS outcomes, companies can move from reactive analysis to proactive strategy. Instead of waiting for negative feedback to surface, predictive modeling can identify pain points and opportunities in real time, allowing marketing and customer service teams to intervene earlier, personalize experiences, and build stronger customer relationships. This project is grounded in this exact

philosophy: using data-driven insights to understand what drives customer loyalty and to help businesses use NPS not just as a metric, but as a catalyst for transformation.

This project seeks to build a random forest model that can accurately estimate a customer's likelihood of promoting or detracting from a fictional bank, based on the characteristics of their complaint and how it was handled. By incorporating variables such as complaint category, origin, language, and time to resolution, this study aims to uncover actionable patterns that drive customer satisfaction and loyalty. Ultimately, this model offers a blueprint for businesses to better anticipate customer needs, prioritize service interventions, and strategically invest in the factors that truly influence NPS ratings. This project hopes to demonstrate how data science can serve as a critical lever for customer-focused marketing and long-term business growth.

3 Methods

3.1 Dataset

The dataset used in this study was obtained from a publicly accessible GitHub repository maintained by Adam Ross Nelson, J.D. Ph.D, a data science teaching professor at the University of Wisconsin - Madison (Nelson, 2024). It contains anonymized records of customer complaints submitted to a fictional bank. The dataset includes 10 variables and 398 observations, each representing a unique customer complaint. Key variables include the bank service associated with the complaint (Product), the language in which the complaint was written (Language), the communication channel used to initiate the complaint (Origin), and the number of days taken to resolve the complaint (Days_To_Resolve). Each observation also includes a free-text complaint (Complaint), a numeric Net Promoter Score (NPS Response) indicating customer satisfaction, and de-identified personal information such as Name and Address, which are not used in the present analysis. This dataset is designed for educational purposes and reflects realistic patterns in customer service interactions within the banking sector.

3.2 Descriptive Statistics

Jupyter Lab was used for the entirety of this data analysis. The dataset was loaded into a Jupyter Notebook directly from Nelson's GitHub Repository and a random sample of three participants was pulled to ensure the datset was read in correctly.

```
# markdown text to table

Markdown(df.sample(3).T.to_markdown())
```

Table 1: Dataset Sample of 3 Customers

	105	91	7
Customer_ID	1106	1092	1008
Product	bill pay	atm	atm
Date	2023-12-16	2023-11-18	2023-06-12
Complaint	Paying bills on your system is like playing a game that I always lose! It never works!	Your ATM has a wonderful habit of eating my cards. Who knew banking could be such an adventure?	The ATMs always seem to be out of service at your uptown branch. This is a recurring issue and needs to be addressed.
Language	English	English	English
Days_To_Resolve	2.7140079981724857	2.2348953956537865	2.1608083023485425
Origin	Telephone	Telephone	Telephone
NPS Response	7	8	8
Name	Ellen Garcia	Shelia Collins	Todd Green

Table 1: Dataset Sample of 3 Customers

	105	91	7
Address	32964 Solomon Field Suite 871 Ramirezview, VI 25561	7795 Orr Junction Suite 338 Lisabury, ND 89842	7971 Cruz Roads Cindychester, FM 56827

The shape of the data frame was checked and revealed a dataset with 10 columns (features) and 398 observations (customers with complaints).

```
# shape of dataset (# of rows, # of columns)
df.shape
```

```
(398, 10)
```

Unique responses and value counts for all three categorical predictors were reviewed. "Product" has six possible unique responses: 'web branch', 'atm', 'checking', 'bill pay', 'credit or debit card', or 'other'. A check on value counts showed a fairly even distribution of responses (~66 counts in each response) with the 'other' category having slightly more counts (71).

```
# unique responses in "Product"
df["Product"].unique()
```

```
array(['web branch', 'atm', 'checking', 'bill pay',
       'credit or debit card', 'other'], dtype=object)
```

```
# value counts for "Product"
df["Product"].value_counts()
```

```
Product
other                  71
web branch             66
atm                    66
checking               66
credit or debit card   66
bill pay               63
Name: count, dtype: int64
```

"Origin" has three possible unique responses: 'Telephone', 'Email', and 'SMS'. Checking the value counts revealed an uneven distribution among responses. 70 comlaints were originated through SMS, 79 through email, and a staggering 249 through telephone. This is a clear indicator that customers prefer to handle their complaints with the bank over the phone as opposed to any other method of communication.

```
# unique responses in "Origin"
df["Origin"].unique()
```

```
array(['Telephone', 'Email', 'SMS'], dtype=object)
```

```
# value counts for "Origin"
df["Origin"].value_counts()
```

```
Origin
Telephone    249
Email         79
SMS           70
Name: count, dtype: int64
```

"Language" has three possible responses: 'English', 'Spanish', and 'French'. A check of value counts revealed the most uneven distribution in this datset, with Spanish and French complaints constituting only 16 counts, each, and English complaints constituting 366 counts.

```
# unique responses in "Language"
df["Language"].unique()
```

```
array(['English', 'Spanish', 'French'], dtype=object)
```

```
# value counts for "Language"
df["Language"].value_counts()
```

```
Language
English    366
Spanish     16
French      16
Name: count, dtype: int64
```

The dataset was checked for missing values and revealed no missing values.

```
# sum of missing cells in whole df
df.isna().sum().sum()
```

```
0
```

Descriptive statistics from the functional bank complaints dataset were obtained, providing a snapshot of the key variables analyzed in this project (see Table 2). The dataset includes 398 customer complaints, with Days_To_Resolve ranging from approximately 0.83 to 2.56 days (M = 1.49, SD = 0.43), indicating relatively swift resolution times across the board. NPS Response, which reflects customers' likelihood of recommending the bank, spans the full 0–10 scale, with an average score of 6.28 (SD = 2.80). The median NPS score is 7, suggesting a slight skew toward promoter responses, although the presence of lower scores indicates notable variability in customer experiences. These initial insights underscore the importance of examining how resolution time and other complaint characteristics may influence overall customer satisfaction and advocacy.

```
# descriptive statistics

df.describe()
```

Table 2: Descriptive Statistics of Numeric Features

	CustomerID	Days_To_Resolve	NPS Response
count	398.000000	398.000000	398.000000
mean	1199.500000	1.488138	6.276382
std	115.036951	0.427025	2.796002
min	1001.000000	0.834867	0.000000
25%	1100.250000	1.120414	4.000000
50%	1199.500000	1.405415	7.000000
75%	1298.750000	1.789266	8.000000
max	1398.000000	2.564467	10.000000

3.3 Exploratory Data Analysis

Exploratory data analysis for this dataset consisted of two components. First, all categorical predictors were plotted against "NPS Response" in violin plots. Violin plots are an effective visualization techniques for categorical predictors that have a continuous outcome. The second step in EDA consisted calculating the

correlation between “NPS Response” and “Days_To_Resolve”, which is the only continuous predictor in this dataset.

The violin plot for “Product” and “NPS Response” shows a very even distribution across responses with a median score of 7-8 across all responses (see Figure 1). This indicates that users of all products had similar NPS responses, and that “Product” may not be a significant predictor of NPS resonse.

```
# EDA - Explore the predictors & their relationship with the outcome...
# categoical violin plot

sns.violinplot(data=df, x = "Product", y = "NPS Response")
```

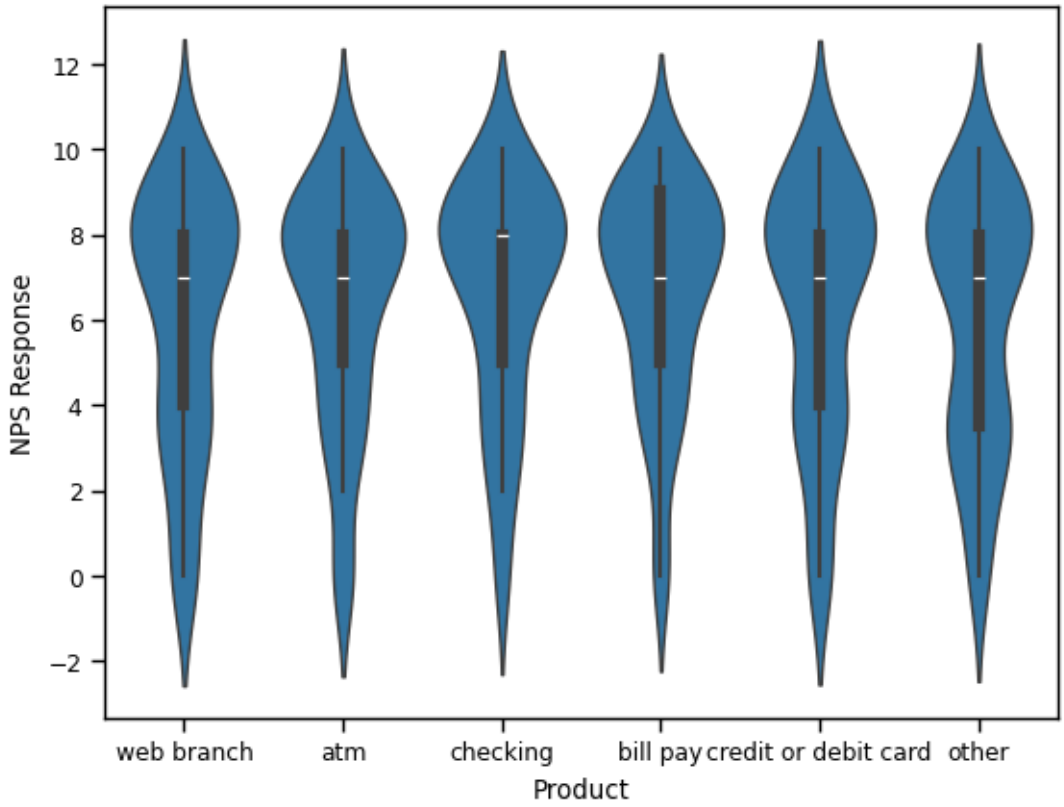


Figure 1: Violin Plot of Product & NPS Response

A violin plot showing the relationship between “Language” and “NPS Response” revealed different distributions across all three responses (see Figure 2). Customers who submitted complaints in English and Spanish were likely to have a higher NPS response (median of 7/8) than customers who submitted their complaints in French (median of 5). This might indicate a discrepancy in the way French complaints are handled, and has a higher chance of being an influential predictor in the random forest model.

```
# relationship between "Language" and "NPS Reponse"

sns.violinplot(data=df, x = "Language", y = "NPS Response")
```

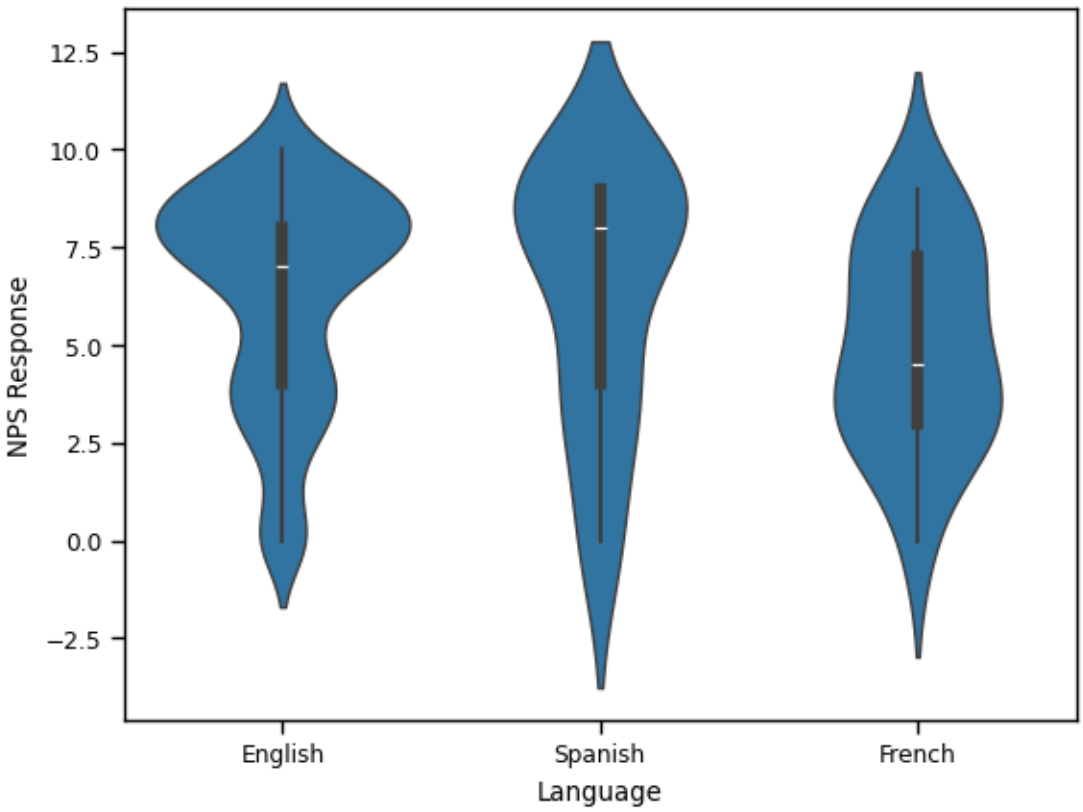


Figure 2: Violin Plot of Language & NPS Response

Similarly to "Product", "Origin" seems to have a weak correlation with "NPS Response" (see Figure 3). Distributions of response counts are even across all responses with a median net promoter score of 7-8. This predictor might be the weakest predictor in the final random forest model due to the lack of variation in distributions.

```
# relationship between "Origin" and "NPS Reponse"

sns.violinplot(data=df, x = "Origin", y = "NPS Response")
```

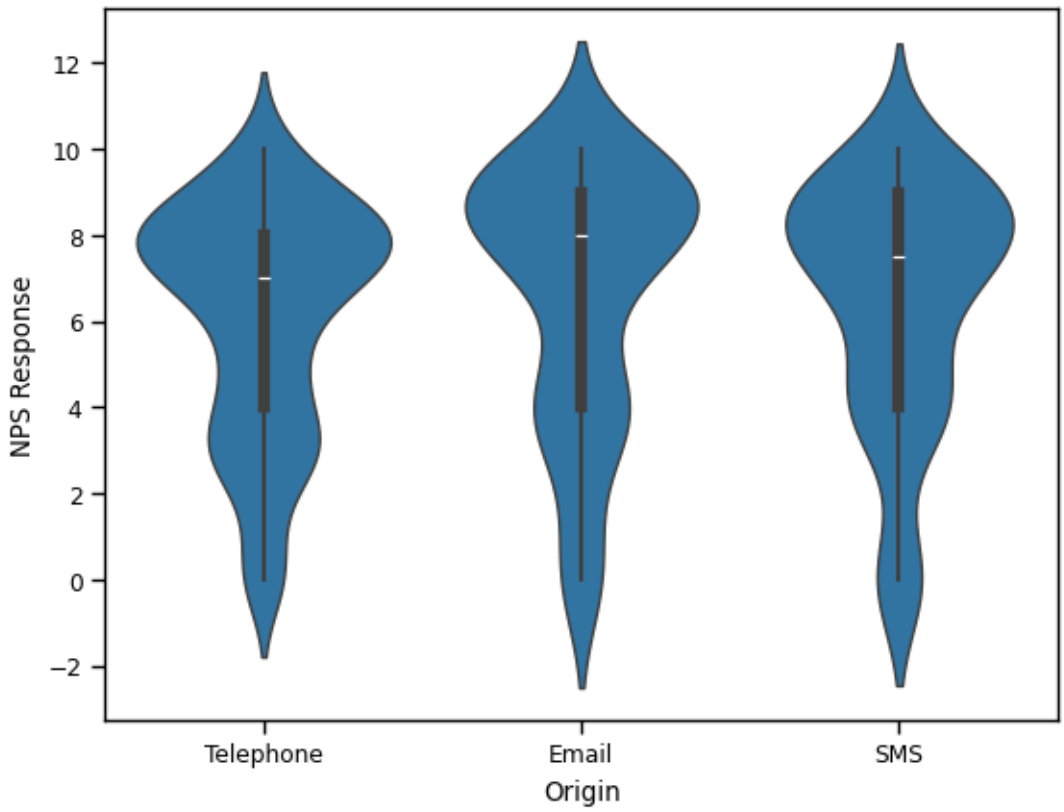


Figure 3: Violin Plot of Origin & NPS Response

A violin plot was created to inspect the relationship between "Days_To_Resolve" and "NPS Response", displayed in Figure 4. "Days_To_Resolve" is a read in as a continuous variable and is treated as such in data analysis. Hor visualization in EDA, this variable performs much better when it is binned into 5 categories than it does as a completely continuous variable. This is because it makes patterns much easier to recognize at a glance by making these patterns more visually robust. "Days_To_Respond" responses range from 0.8 days to 2.6 days, which were binned into three groups.

This violin plot revealed similar patterns across almost all resolution bins. Complaints that were resolved within 1.4 days have a median NPS response of 8, while complaints that took longer than 1.4 days have a median NPS response of 7. This might suggest that customers whose complaints are resolved quicker are more likely to recommend this bank to their friends.

```
# relationship between "Days_To_Resolve" and "NPS Response"

df["days_binned"] = pd.cut(df["Days_To_Resolve"], bins=3)
sns.violinplot(data=df, x="days_binned", y="NPS Response")
```

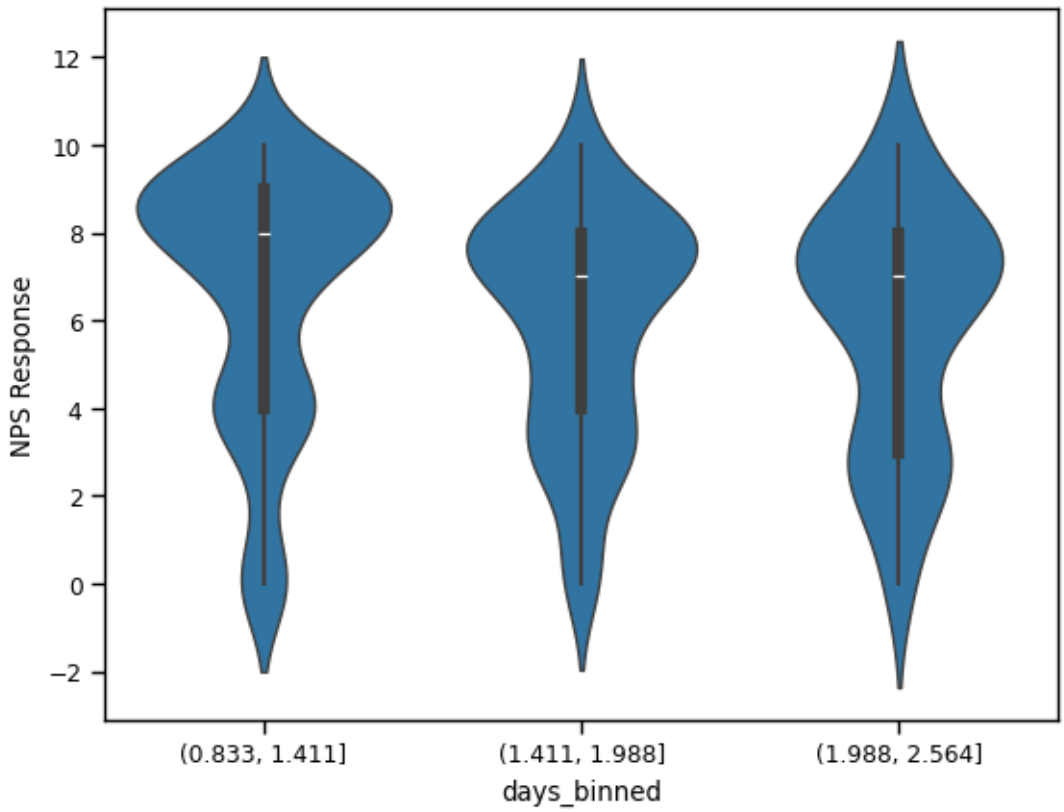


Figure 4: Violin Plot of days_binned & NPS Response

The final step in exploratory data analysis involved calculating the correlation coefficient for "Days_To_Resolve" and "NPS Response", displayed in Table 3. This calculation revealed a correlation effect of -0.10, indicating a relatively weak negative correlation between days taken to resolve a complaint and NPS response.

```
# correlation between "Days_To_Resolve" & "NPS Response"

df[['Days_To_Resolve','NPS Response']].corr()
```

Table 3: Correlation between Days_To_Resolve & NPS Response

	Days_To_Resolve	NPS Response
Days_To_Resolve	1.000000	-0.101308
NPS Response	-0.101308	1.000000

3.4 Random Forest Model

3.4.1 Random Forest with One-Hot Encoding

A random forest model was run in this project using OneHotEncoder(). The categorical predictors were all combined into one Python list for easier use later in the model. The outcome variable was defined as "NPS Response". The numeric variable Days_To_Resolve was standardized using StandardScaler() from sklearn, ensuring it has a mean of 0 and standard deviation of 1, which helps the model treat it appropriately during training. Next, the categorical variables ("Language", "Product", and "Origin") were one-hot encoded using OneHotEncoder, converting each category into binary columns so they can be used in the model. These

processed numeric and categorical features were then concatenated into a single dataset X_1. The data were split into training and testing sets using a 67%-33% ratio. A random forest regression model (rf_1) was then trained on the training data. Finally, predictions were made on the test set, and the model's performance was evaluated using Root Mean Squared Error (RMSE).

```
# random forest model with one-hot encoding

# define predictors
cat_preds = ['Language', 'Product', 'Origin']
y_1 = df['NPS Response']

# scale numeric feature
the_scaler = StandardScaler().set_output(transform="pandas")
X_num_1 = the_scaler.fit_transform(df[["Days_To_Resolve"]])

# One-hot encode categorical features
encoder = OneHotEncoder(sparse_output=False, handle_unknown="ignore")
x_cat_array_1 = encoder.fit_transform(df[cat_preds])
X_cat_1 = pd.DataFrame(x_cat_array_1,
                       columns=encoder.get_feature_names_out(cat_preds),
                       index=df.index)

# combine all features into X
X_1 = pd.concat([X_num_1, X_cat_1], axis=1)

# train-test split
X_train_1, X_test_1, y_train_1, y_test_1 = train_test_split(
    X_1, y_1,test_size=0.33, random_state=42)

# random Forest Regression
rf_1 = RandomForestRegressor(random_state=42)
rf_1.fit(X_train_1, y_train_1)

# RMSE
rmse_1 = np.sqrt(mean_squared_error(y_test_1, rf_1.predict(X_test_1)))
print("RMSE (Model 1):", rmse_1)
```

```
RMSE (Model 1): 3.3518035624439744
```

3.5 Feature Importance

To better understand which variables were most influential in predicting net promoter score, feature importance was extracted from the trained random forest model. A data frame was created that paired each feature name from the training set with its corresponding importance score as calculated by the model (Table 4). These importance values reflected the relative influence of each predictor in the model's performance, overall. A horizontal bar plot was then generated using Seaborn's barplot() function to visually display these feature importances, with the features ordered along the y-axis and their corresponding importance values on the x-axis (**?@fig-rf1**).

```
# feature importance for rf_1

viz_fi_1 = pd.DataFrame({'Features': X_train_1.columns, 'Importance':
                         rf_1.feature_importances_})
viz_fi_1.style.hide()
```

Table 4: Feature Importance for Random Forest Model 1

Table 4

Features	Importance
Days_To_Resolve	0.673872
Language_English	0.034767
Language_French	0.007704
Language_Spanish	0.011402
Product_atm	0.022341
Product_bill pay	0.029445
Product_checking	0.025389
Product_credit or debit card	0.038802
Product_other	0.037535
Product_web branch	0.037379
Origin_Email	0.023077
Origin_SMS	0.034029
Origin_Telephone	0.024257

```
# feature importance bar plot (rf_1)

sns.barplot(viz_fi_1, x = 'Importance', y = 'Features')
```

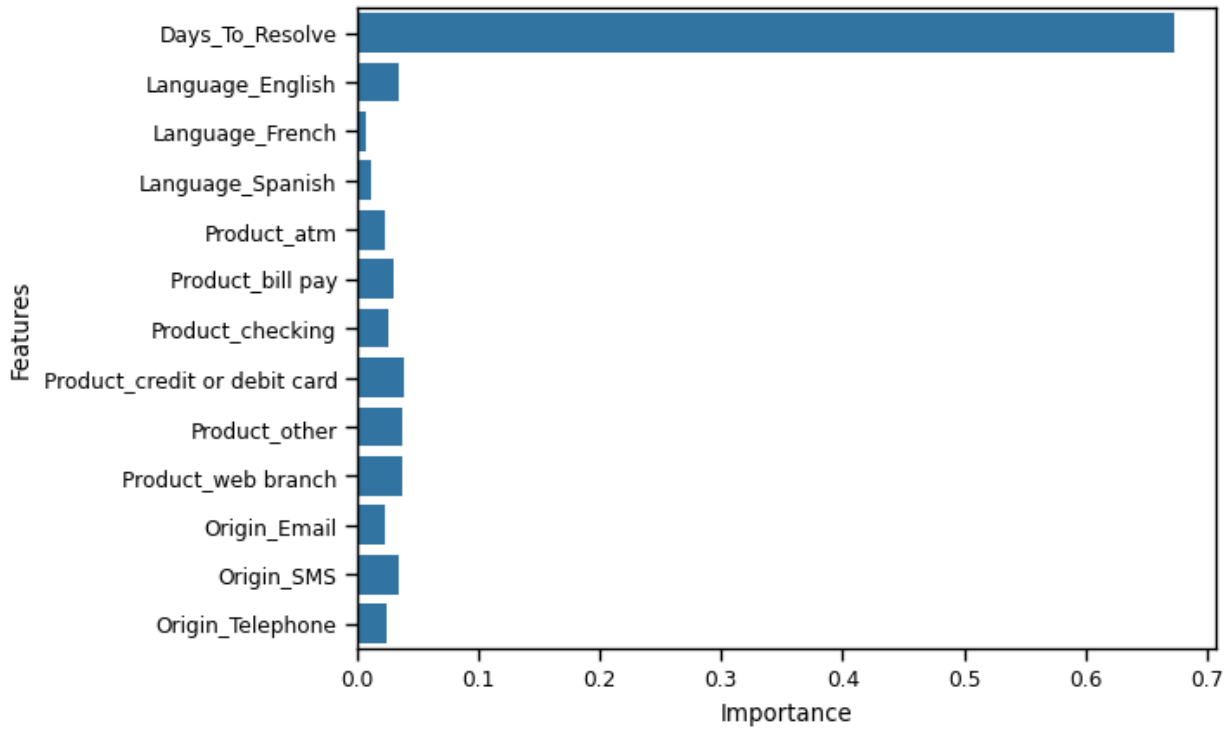


Figure 5: Horizontal Bar Plot of Feature Importance for Random Forest Model 1

4 Results

4.1 Model Performance

The random forest model was evaluated based on its Root Mean Squared Error (RMSE) on the test data. Using OneHotEncoder(), the model achieved an RMSE of 3.52. The bar plot visualizations make it easier to interpret which variables had the greatest impact on NPS prediction, clearly showing that the number of days taken to resolve complaint had the greatest influence on the customer's reported net promoter score. Additionally, the barplot shows that "French" and "Spanish" complaints had the least influence in the model, which is consistent with there being only 16 observations in each response category.

4.2 Feature Importance

To identify the most influential predictors of NPS, feature importance values were extracted from the trained random forest model. Days_To_Resolve emerged as the most important predictor by a significant margin (see **?@fig-rf1**). This result aligns with earlier findings from exploratory data analysis, which showed a small but meaningful negative correlation between resolution time and NPS response in **?@tbl-crr** (r = -0.10). Shorter resolution times were associated with higher NPS scores, indicating that customers value swift complaint resolution.

Among the categorical features, Language had higher importance than Product and Origin, with complaints in French associated with lower NPS scores. However, their influence was substantially lower than that of Days_To_Resolve. Product and Origin had the lowest importance scores, reflecting their relatively uniform NPS distributions observed during exploratory visualization.

The consistent prominence of Days_To_Resolve in both models underscores the critical role of efficient issue resolution in shaping customer loyalty and advocacy. The feature importance analysis confirms that resolution speed is a key driver of customer satisfaction and should be prioritized in service improvement efforts.

5 Discussion

The goal of this analysis was to identify complaint characteristics that best predict customers' Net Promoter Scores (NPS) in order to inform strategies for improving customer satisfaction. Using random forest regression, Days_To_Resolve was found to be the most important predictor of NPS, outperforming all categorical variables by a substantial margin. This finding supports the hypothesis that resolution speed is a critical componenet of customer sentiment.

The strong importance of Days_To_Resolve is consistent with prior research and business intuition: customers are more likely to rate their experience positively when their issues are resolved quickly. Although the correlation between Days_To_Resolve and NPS was modest (r = -0.10), the feature's central role in the model emphasizes its practical relevance when considered alongside other predictors.

Among categorical features, Language had the next highest importance, with complaints in French associated with slightly lower NPS. While Product and Origin offered minimal additional predictive value, their inclusion may still help capture small subgroup differences or interaction effects in future modeling efforts.

Overall, this analysis underscores the business impact of timely resolution. Organizations seeking to improve customer satisfaction should consider investing in tools, staffing, or workflows that reduce the time needed to resolve complaints. Reducing resolution time may yield disproportionately positive effects on customer perception and loyalty, as reflected in higher NPS ratings.

To build on these findings, future work could explore incorporating sentiment analysis of the complaint text as a predictor of NPS. Sentiment scores may capture nuances in customer emotion that are not fully explained by structural features like resolution time or complaint category. This could enhance model performance and offer deeper insights into what drives customer loyalty.

In addition to sentiment, segmenting the data by customer type or complaint severity may uncover subgroup patterns that may not emerge in aggregate models, enabling more targeted strategies for improving satisfaction. Ultimately, improving customer satisfaction isn't just about better service; it's a core element of long-term business growth.

6 Appendix

```
# data dictionary

data_dict = pd.DataFrame({
    "Feature": [
        "Product", "Language", "Origin", "Days_To_Resolve", "Complaint", "NPS_Response"
    ],
    "Explanation": [
        "Bank service used",
        "Language of complaint",
        "Origin of complaint",
        "Number of days taken to resolve complaint",
        "Text data of complaints",
        "Net Promoter Score"
    ],
    "Variable Type": [
        "Categorical", "Categorical", "Categorical", "Continuous", "Text", "Continuous"
    ],
    "Unit of Measurement": [
        "ATM, Web Branch, Bill Pay, Checking, Credit/Debit Card, Other",
        "English, Spanish, French",
        "Telephone, Email, SMS",
        "0 to 3 days",
        "Free text (varied length)",
        "0 to 10"
    ]
})

data_dict.style.hide()
```

Table 5: Data Dictionary for Fictional Bank Compaints Dataset

Table 5

Feature	Explanation	Variable Type	Unit of Measurement
Product	Bank service used	Categorical	ATM, Web Branch, Bill Pay, Checking,
Language	Language of complaint	Categorical	English, Spanish, French
Origin	Origin of complaint	Categorical	Telephone, Email, SMS
Days_To_Resolve	Number of days taken to resolve complaint	Continuous	0 to 3 days
Complaint	Text data of complaints	Text	Free text (varied length)
NPS_Response	Net Promoter Score	Continuous	0 to 10

7 References

1. Dawes, J. G. (2023). The net promoter score: What should managers know? International Journal of Market Research, 66(2-3), 182-198. https://doi.org/10.1177/14707853231195003

2. Markey, R., Reichheld, F., & Dullweber, A. (2020). Are You Undervaluing Your Customers? Harvard Business Review, 98(1), 60–67.

3. Reichheld, F. F. (2003). The one number you need to grow. Harvard Business Review, 81(12), 46–54.

4. https://www.linkedin.com/posts/nickhr_nps-what-should-managers-know-activity-7221955116851965952-PeKG/

5. Above is a linkedin url to article #1 in my References list. Someone uploaded a sketchy pdf of this journal article on their linkedin page and for some reason this is the only place I could find the article for free. I am keeping the url in here for easy referencing.